

# Automating Infrastructure Deployment with AWS CloudFormation

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AWS CloudFormation

# Important Disclaimer

- **No Endorsement:** This guide is for educational purposes only and does not imply endorsement by Metabase or AWS.
- **Testing Only:** Intended for testing and development, not production environments.
- **Refer to Official Docs:** Always consult official AWS documentation for production use.
- **Proceed at Your Own Risk:** Use this guide at your own risk and be aware of potential deployment implications.

# Overview of AWS CloudFormation



## **AWS CloudFormation**

AWS CloudFormation is a service that lets you define and provision AWS resources using templates, so you can focus on your applications instead of manually managing infrastructure. You write a template describing the resources you need, such as Amazon EC2 instances or Amazon RDS databases, and CloudFormation automatically provisions and configures them in the correct order, handling all dependencies for you.

# Business Case for Infrastructure Automation



## **AWS CloudFormation**

AWS CloudFormation lets you define AWS resources as code, enabling fast, consistent, and error-free deployments.

This project uses templates to automate the network layer (VPC and subnet) and the application layer (EC2 instance and security group), ensuring the application setup automatically connects to the network.

The approach reduces human errors, saves time, and simplifies updates or scaling across different environments like development and production.

# Designing the Network YAML Template

```
AWSTemplateFormatVersion: 2010-09-09
Description: >-
  Network Template: Sample template that creates a VPC with DNS and public IPs enabled.

# This template creates:
#   VPC
#   Internet Gateway
#   Public Route Table
#   Public Subnet

#####
# Resources section
#####

Resources:

  ## VPC

  VPC:
    Type: AWS::EC2::VPC
    Properties:
      EnableDnsSupport: true
      EnableDnsHostnames: true
      CidrBlock: 10.0.0.0/16

  ## Internet Gateway

  InternetGateway:
    Type: AWS::EC2::InternetGateway
```

Draft and validate network.yaml to define the VPC, public/private subnets, and related networking resources. Make sure outputs such as VPC ID and Public Subnet IDs are exported for use by other stacks.

For a more complete example, visit this GitHub repo: [\[https://github.com/ikhsannur1996/cloud-formation\]](https://github.com/ikhsannur1996/cloud-formation).

# Initiating the Networking Stack Creation

aws Search [Alt+S] United States (N. Virginia)

CloudFormation > Stacks > Create stack

Step 1: Create stack (selected)  
Step 2: Specify stack details  
Step 3: Configure stack options  
Step 4: Review and create

### Create stack

**Prerequisite - Prepare template**  
You can also create a template by scanning your existing resources in the [laC generator](#).

**Prepare template**  
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Choose an existing template  
Upload or choose an existing template.

☐ Build from Infrastructure Composer  
Create a template using a visual builder.

**Specify template** [Info](#)  
This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

**Template source**  
Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

☐ Amazon S3 URL  
Provide an Amazon S3 URL to your template.

☒ Upload a template file  
Upload your template directly to the console.

☐ Sync from Git  
Sync a template from your Git repository.

**Upload a template file**

[Choose file](#)

network.yaml

JSON or YAML, formatted file

S3 URL: <https://s3.us-east-1.amazonaws.com/cf-templates-5th4y6Soa140-us-east-1/2025-09-21T110301.969Zhmp-network.yaml> [View in Infrastructure Composer](#)

[Cancel](#) [Next](#)

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Go to AWS CloudFormation and select Create stack → With new resources (standard).  
Upload the template file network.yaml to start creating the networking layer stack.

# Defining Networking Stack Parameters

Step 1  
Create stack

Step 2  
**Specify stack details**

Step 3  
Configure stack options

Step 4  
Review and create

### Specify stack details

**Provide a stack name**

Stack name

ikhsan-network

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 14/128.

**Parameters**

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

**No parameters**

There are no parameters defined in your template

Cancel Previous Next

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Enter the Stack name as ikhsan-network to identify the networking stack.  
No additional parameters are required for this stack.

# Configuring Networking Stack Options

The screenshot shows the AWS CloudFormation console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and a location dropdown set to 'United States (N. Virginia)'. Below the navigation bar, the breadcrumb trail reads 'CloudFormation > Stacks > Create stack'. On the left side, a vertical progress bar indicates four steps: 'Step 1: Create stack', 'Step 2: Specify stack details', 'Step 3: Configure stack options' (which is the current step and highlighted with a blue circle), and 'Step 4: Review and create'.

The main content area is titled 'Configure stack options'. It contains three sections:

- Tags - optional**: A section explaining that tags are key-value pairs used for metadata. It includes a table with two columns: 'Key' and 'Value - Tags - optional'. The first row shows 'application' as the key and 'incubator-program' as the value. There are 'Add new tag' and 'Remove' buttons. A note at the bottom says 'You can add 49 more tag(s)'.
- Permissions - optional**: A section explaining that you can specify an existing IAM role. It includes a dropdown for 'IAM role name' with 'Sample-role-name' selected, and a 'Remove' button.
- Stack failure options**: A section titled 'Behavior on provisioning failure' with a 'Learn more' link. It has two radio button options: 'Roll back all stack resources' (which is selected) and 'Preserve successfully provisioned resources'.

At the bottom of the console, there's a footer with 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Leave the default settings or add tags and permissions if needed for better resource management.



# Review and Launch Networking Stack

The screenshot shows the AWS CloudFormation console interface. At the top, the AWS logo and navigation bar are visible, including a search bar and the region 'United States (N. Virginia)'. Below the navigation bar, the breadcrumb trail reads 'CloudFormation > Stacks > Create stack'. On the left side, a vertical progress bar indicates four steps: 'Step 1: Create stack', 'Step 2: Specify stack details', 'Step 3: Configure stack options', and 'Step 4: Review and create'. The 'Review and create' step is currently selected and highlighted with a blue circle. The main content area is titled 'Review and create' and contains two sections. The first section, 'Step 1: Specify template', includes a 'Prerequisite - Prepare template' box stating 'Template is ready', a 'Template' box showing the 'Template URL' as 'https://s3.us-east-1.amazonaws.com/cf-templates-5th4y6S0a140-us-east-1/2025-09-21T110301.969Zhmp-network.yaml' and a 'Stack description' as 'Network Template: Sample template that creates a VPC with DNS and public IPs enabled.', and an 'Edit' button. The second section, 'Step 2: Specify stack details', includes a 'Provide a stack name' box with the 'Stack name' set to 'ikhsan-network' and an 'Edit' button. Below this is a 'Parameters' section with a search bar. At the bottom of the console, the footer shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

aws Search [Alt+S] United States (N. Virginia)

CloudFormation > Stacks > Create stack

Step 1  
Create stack

Step 2  
Specify stack details

Step 3  
Configure stack options

Step 4  
Review and create

## Review and create

### Step 1: Specify template

**Prerequisite - Prepare template**

**Template**  
Template is ready

**Template**

**Template URL**  
https://s3.us-east-1.amazonaws.com/cf-templates-5th4y6S0a140-us-east-1/2025-09-21T110301.969Zhmp-network.yaml

**Stack description**  
Network Template: Sample template that creates a VPC with DNS and public IPs enabled.

### Step 2: Specify stack details

**Provide a stack name**

**Stack name**  
ikhsan-network

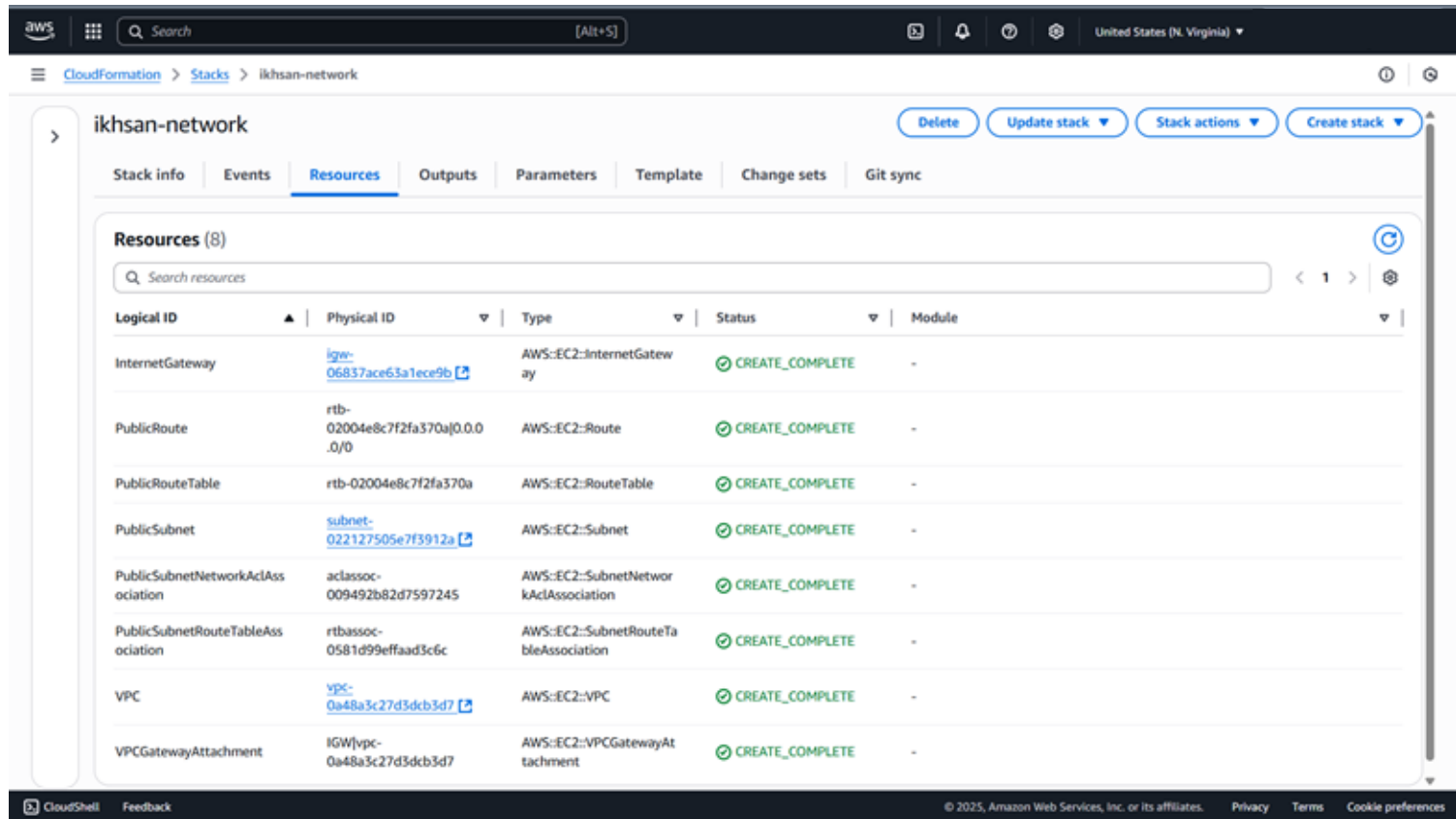
**Parameters**

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Check the template details and configuration, then click Create stack.

Wait until the stack status shows CREATE\_COMPLETE, confirming the networking resources have been deployed.

# Validating Deployed Network Resources



The screenshot displays the AWS CloudFormation console interface. At the top, the navigation bar shows the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The breadcrumb trail indicates the path: CloudFormation > Stacks > ikhsan-network. The main header for the stack 'ikhsan-network' includes buttons for 'Delete', 'Update stack', 'Stack actions', and 'Create stack'. Below this, a tabbed interface shows 'Stack info', 'Events', 'Resources' (selected), 'Outputs', 'Parameters', 'Template', 'Change sets', and 'Git sync'. The 'Resources' tab displays a list of 8 resources, each with a logical ID, physical ID, type, status, and module. All resources are in a 'CREATE\_COMPLETE' state.

| Logical ID                        | Physical ID                              | Type                                  | Status          | Module |
|-----------------------------------|------------------------------------------|---------------------------------------|-----------------|--------|
| InternetGateway                   | <a href="#">igw-06837ace63a1ece9b</a>    | AWS::EC2::InternetGateway             | CREATE_COMPLETE | -      |
| PublicRoute                       | rtb-02004e8c7f2fa370a 0.0.0.0/0          | AWS::EC2::Route                       | CREATE_COMPLETE | -      |
| PublicRouteTable                  | rtb-02004e8c7f2fa370a                    | AWS::EC2::RouteTable                  | CREATE_COMPLETE | -      |
| PublicSubnet                      | <a href="#">subnet-022127505e7f3912a</a> | AWS::EC2::Subnet                      | CREATE_COMPLETE | -      |
| PublicSubnetNetworkACLAssociation | aclassoc-009492b82d7597245               | AWS::EC2::SubnetNetworkACLAssociation | CREATE_COMPLETE | -      |
| PublicSubnetRouteTableAssociation | rtbassoc-0581d99effaad3c6c               | AWS::EC2::SubnetRouteTableAssociation | CREATE_COMPLETE | -      |
| VPC                               | <a href="#">vpc-0a48a3c27d3dcb3d7</a>    | AWS::EC2::VPC                         | CREATE_COMPLETE | -      |
| VPCGatewayAttachment              | IGW[vpc-0a48a3c27d3dcb3d7                | AWS::EC2::VPCGatewayAttachment        | CREATE_COMPLETE | -      |

Open the Resources tab to view the created VPC, subnets, and related networking components. Check the Outputs tab to find the exported VPC ID and Public Subnet IDs for use by other stacks.

# Designing the Application YAML Template

```

AWS::CloudFormation::Template
AWSTemplateFormatVersion: 2010-09-09
Description: >-
  Application Template: Demonstrates how to reference resources from a different stack.
  This template provisions an EC2 instance in a VPC Subnet provisioned in a different stack.

# This template creates:
#   Amazon EC2 instance
#   Security Group

#####
# Parameters section
#####

Parameters:

  NetworkStackName:
    Description: >-
      Name of an active CloudFormation stack that contains the networking
      resources, such as the VPC and subnet that will be used in this stack.
    Type: String
    MinLength: 1
    MaxLength: 255
    AllowedPattern: '^[a-zA-Z][-a-zA-Z0-9]*$'
    Default: ikhsan-network

  AmazonLinuxAMIID:
    Type: AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>
    Default: /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2
```

Draft and validate application.yaml to deploy the EC2 instance and security group while referencing outputs from the network stack. Ensure the parameter NetworkStackName is set to import VPC and subnet IDs from the network stack outputs.

For a complete reference setup and template examples, visit this GitHub repo: [<https://github.com/ikhsannur1996/cloud-formation>].

# Launching the Application Stack Creation

**Create stack**

**Step 1: Create stack**

**Prerequisite - Prepare template**

You can also create a template by scanning your existing resources in the [IaC generator](#).

**Prepare template**

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Choose an existing template  
Upload or choose an existing template.

☐ Build from Infrastructure Composer  
Create a template using a visual builder.

**Specify template** [info](#)

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

**Template source**

Selecting a template generates an Amazon S3 URL, where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

☐ Amazon S3 URL  
Provide an Amazon S3 URL to your template.

☒ Upload a template file  
Upload your template directly to the console.

☐ Sync from Git  
Sync a template from your Git repository.

**Upload a template file**

[Choose file](#)

application.yaml

JSON or YAML, formatted file

S3 URL: <https://s3.us-east-1.amazonaws.com/cf-templates-5th4y6Soa140-us-east-1/2025-09-21T110618.7412dc8-application.yaml> [View in Infrastructure Composer](#)

[Cancel](#) [Next](#)

Go to AWS CloudFormation and select Create stack → With new resources (standard).  
Upload the template file application.yaml to start creating the application layer stack.

# Defining Application Stack Parameters

The screenshot shows the AWS CloudFormation console interface. At the top, the navigation bar includes the AWS logo, a search bar, and account information (United States (N. Virginia), Account ID: 7976-6678-1058, Gye Goon Woo). The breadcrumb trail indicates the path: CloudFormation > Stacks > Create stack. On the left, a vertical progress bar shows four steps: Step 1: Create stack, Step 2: Specify stack details (highlighted with a blue circle), Step 3: Configure stack options, and Step 4: Review and create. The main content area is titled 'Specify stack details' and contains three sections: 1. 'Provide a stack name' with a text input field containing 'ikhsan-application' and a note: 'Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 18/128.' 2. 'Parameters' with a sub-header 'Parameters are defined in your template and allow you to input custom values when you create or update a stack.' 3. Two parameter input fields: 'AmazonLinuxAMIID' with the value '/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86\_64-gp2' and 'NetworkStackName' with the value 'ikhsan-network'. At the bottom right of the form are three buttons: 'Cancel', 'Previous' (highlighted with a blue border), and 'Next' (highlighted with an orange background). The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Step 1  
Create stack

Step 2  
**Specify stack details**

Step 3  
Configure stack options

Step 4  
Review and create

### Specify stack details

**Provide a stack name**

**Stack name**

ikhsan-application

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 18/128.

**Parameters**

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

**AmazonLinuxAMIID**

/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86\_64-gp2

**NetworkStackName**

Name of the CloudFormation stack that created the networking resources (VPC & subnet) used by this stack.

ikhsan-network

Cancel Previous Next

Enter the Stack name as ikhsan-application. Set the parameter NetworkStackName to ikhsan-network so the application stack can import the VPC and subnet IDs from the networking stack outputs.

# Configuring Application Stack Options

The screenshot shows the AWS CloudFormation console interface. On the left, a navigation pane lists four steps: 'Step 1: Create stack', 'Step 2: Specify stack details', 'Step 3: Configure stack options' (which is selected and highlighted with a blue circle), and 'Step 4: Review and create'. The main content area is titled 'Configure stack options' and contains three sections: 'Tags - optional', 'Permissions - optional', and 'Stack failure options'. The 'Tags' section has a table with two columns: 'Key' and 'Value'. The first row contains 'application' under 'Key' and 'trainer-incubation' under 'Value'. There is an 'Add new tag' button and a note 'You can add 40 more tag(s)'. The 'Permissions' section has a dropdown for 'IAM role name' with 'Sample-role-name' selected, and a 'Remove' button. The 'Stack failure options' section has two radio buttons: 'Roll back all stack resources' (selected) and 'Preserve successfully provisioned resources'. The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates.

aws Search [Alt+S] United States (N. Virginia)

CloudFormation > Stacks > Create stack

Step 1: Create stack  
Step 2: Specify stack details  
Step 3: **Configure stack options**  
Step 4: Review and create

### Configure stack options

#### Tags - optional

Tags (key-value pairs) are used to apply metadata to AWS resources, which can help in organizing, identifying, and categorizing those resources. You can add up to 50 unique tags for each stack.

| Key         | Value - Tags - optional |        |
|-------------|-------------------------|--------|
| application | trainer-incubation      | Remove |

Add new tag

You can add 40 more tag(s)

#### Permissions - optional

Specify an existing AWS Identity and Access Management (IAM) service role that CloudFormation can assume.

**IAM role - optional**

Choose the IAM role for CloudFormation to use for all operations performed on the stack.

IAM role name Sample-role-name Remove

#### Stack failure options

**Behavior on provisioning failure**

Specify the roll back behavior for a stack failure. [Learn more](#)

☒ Roll back all stack resources

Roll back the stack to the last known stable state.

☐ Preserve successfully provisioned resources

Preserves the state of successfully provisioned resources, while rolling back failed resources to the last known stable state. Resources without a last known stable state will be deleted upon the next stack operation.

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Leave the default settings or add tags and permissions as needed for easy identification and access control.

# Review and Deploy Application Stack

The screenshot shows the AWS CloudFormation console interface. At the top, the navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. Below the navigation bar, the breadcrumb trail reads 'CloudFormation > Stacks > Create stack'. On the left side, a vertical list of steps is shown: 'Step 1: Create stack', 'Step 2: Specify stack details', 'Step 3: Configure stack options', and 'Step 4: Review and create'. The 'Review and create' step is currently selected and highlighted with a blue circle. The main content area is titled 'Review and create' and contains two sections: 'Step 1: Specify template' and 'Step 2: Specify stack details'. The 'Step 1: Specify template' section includes a 'Prerequisite - Prepare template' box stating 'Template is ready', and a 'Template' box showing the 'Template URL' as 'https://s3.us-east-1.amazonaws.com/cf-templates-5th4y65oa140-us-east-1/2025-09-21T111248.995Zafq-application.yaml' and a 'Stack description' that reads: 'Application Template: Demonstrates how to reference resources from a different stack. This template provisions an EC2 instance in a VPC Subnet provisioned in a different stack, downloads index.html from S3, and serves it using Apache.' The 'Step 2: Specify stack details' section includes a 'Provide a stack name' box with the 'Stack name' set to 'ikhsan-application', and a 'Parameters (2)' box. Each section has an 'Edit' button in the top right corner. The bottom of the console shows a 'CloudShell' button, a 'Feedback' link, and a footer with copyright information and links for 'Privacy', 'Terms', and 'Cookie preferences'.

Step 1  
Create stack

Step 2  
Specify stack details

Step 3  
Configure stack options

Step 4  
Review and create

## Review and create

### Step 1: Specify template

**Prerequisite - Prepare template**

**Template**  
Template is ready

**Template**

**Template URL**  
`https://s3.us-east-1.amazonaws.com/cf-templates-5th4y65oa140-us-east-1/2025-09-21T111248.995Zafq-application.yaml`

**Stack description**  
Application Template: Demonstrates how to reference resources from a different stack. This template provisions an EC2 instance in a VPC Subnet provisioned in a different stack, downloads index.html from S3, and serves it using Apache.

### Step 2: Specify stack details

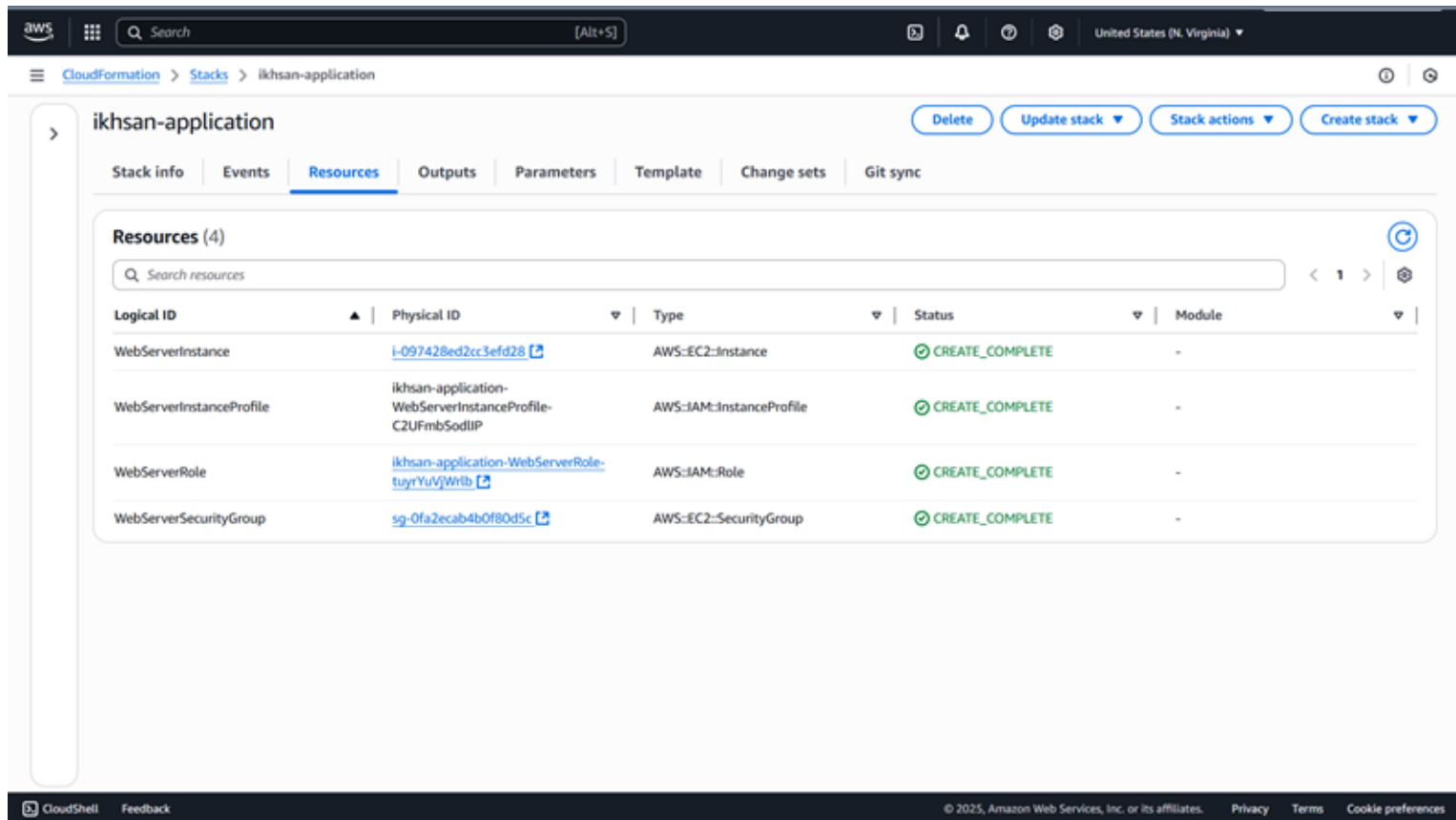
**Provide a stack name**

**Stack name**  
ikhsan-application

**Parameters (2)**

Review the configuration and click Create stack. Wait until the status becomes `CREATE_COMPLETE`, indicating that the EC2 instance and security group have been successfully deployed.

# Verifying Application Resources Deployment



The screenshot displays the AWS CloudFormation console interface. At the top, the navigation bar shows the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The breadcrumb trail indicates the path: CloudFormation > Stacks > ikhsan-application. The main header for the stack 'ikhsan-application' includes buttons for 'Delete', 'Update stack', 'Stack actions', and 'Create stack'. Below this, a tabbed interface shows 'Stack info', 'Events', 'Resources' (selected), 'Outputs', 'Parameters', 'Template', 'Change sets', and 'Git sync'. The 'Resources' tab displays a table of resources.

**Resources (4)**

| Logical ID               | Physical ID                                                  | Type                      | Status          | Module |
|--------------------------|--------------------------------------------------------------|---------------------------|-----------------|--------|
| WebServerInstance        | <a href="#">i-097428ed2cc3efd28</a>                          | AWS::EC2::Instance        | CREATE_COMPLETE | -      |
| WebServerInstanceProfile | ikhsan-application-WebServerInstanceProfile-C2UFmbSodlIP     | AWS::IAM::InstanceProfile | CREATE_COMPLETE | -      |
| WebServerRole            | <a href="#">ikhsan-application-WebServerRole-tuyrYuVjWtB</a> | AWS::IAM::Role            | CREATE_COMPLETE | -      |
| WebServerSecurityGroup   | <a href="#">sg-0fa2ecab4b0f80d5c</a>                         | AWS::EC2::SecurityGroup   | CREATE_COMPLETE | -      |

Open the Resources tab to confirm the EC2 instance and security group are successfully created.



# Accessing the Application Stack Output

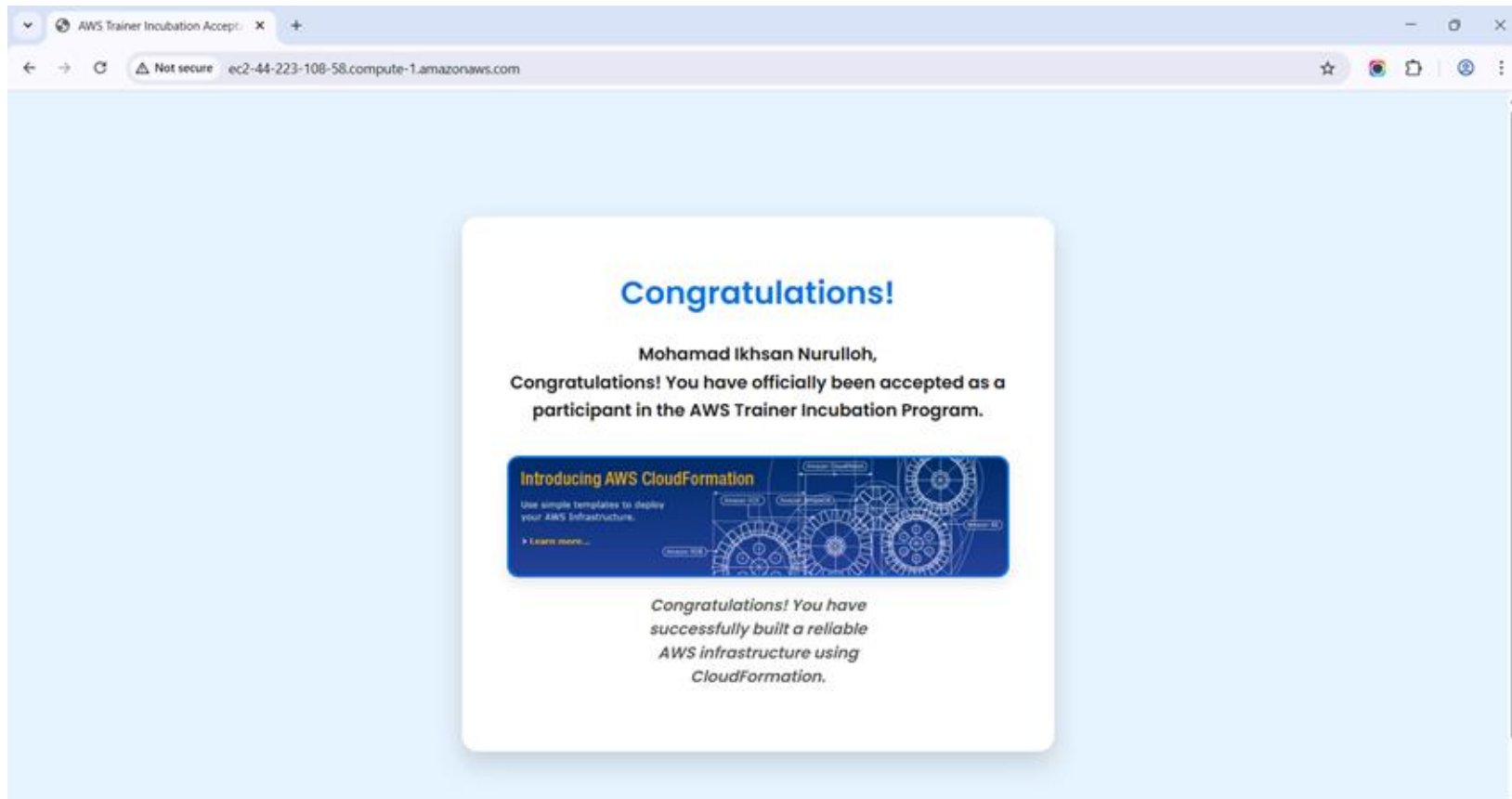
The screenshot shows the AWS CloudFormation console interface. At the top, the navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. Below the navigation bar, the breadcrumb trail reads 'CloudFormation > Stacks > ikhsan-application'. The main content area is titled 'ikhsan-application' and features several action buttons: 'Delete', 'Update stack', 'Stack actions', and 'Create stack'. A horizontal tab bar below the title includes 'Stack info', 'Events', 'Resources', 'Outputs' (which is selected and highlighted), 'Parameters', 'Template', 'Change sets', and 'Git sync'. The 'Outputs' tab displays a section titled 'Outputs (1)' with a search bar labeled 'Search outputs'. Below this is a table with the following data:

| Key | Value                                                                                                           | Description               | Export name |
|-----|-----------------------------------------------------------------------------------------------------------------|---------------------------|-------------|
| URL | <a href="http://ec2-44-200-18-111.compute-1.amazonaws.com">http://ec2-44-200-18-111.compute-1.amazonaws.com</a> | URL of the sample website | -           |

The footer of the console shows 'CloudShell', a 'Feedback' link, and copyright information: '© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

Check the Outputs tab for the application URL, then open it in a browser to verify the application is running on the EC2 instance.

# Testing the Deployed Web Application



Open this URL in a web browser to confirm the application is accessible and running correctly on the deployed EC2 instance.

# Key Outcomes and Observations

- CloudFormation templates successfully automated the creation of network (ikhsan-network) and application (ikhsan-application) stacks.
- Networking outputs (VPC & Subnet IDs) were reused by the application stack, ensuring clean and consistent infrastructure deployment.
- The application URL in Outputs confirmed the EC2 instance was deployed and running as expected.

# Strategic Recommendations for Production

- Store and version CloudFormation templates in Git for easier updates and rollback.
- Integrate with CI/CD pipelines to automate future deployments.
- Add parameters and outputs for flexibility and reusability across environments.
- For production, enhance the template with Auto Scaling, ALB, and CloudWatch monitoring.

Thank You